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**B.PHARM**  
**(SEM I) THEORY EXAMINATION 2022-23**  
**PHARMACEUTICAL CHEMISTRY-I**  
**(PHARMACEUTICAL INORGANIC CHEMISTRY)**

**Time: 3 Hours****Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 7 = 14**

- (a) What do you mean by impurity?
- (b) Classify methods used for limit test of heavy metals.
- (c) What are the advantages of Povidone Iodine solution?
- (d) What do you mean by dentifrices? Write two examples.
- (e) Why combination antacids are required?
- (f) Deduce the composition of ORS.
- (g) Mention the units of radioactivity.

**SECTION B****2. Attempt any three of the following;****7 x 3 = 21**

- (a) Elaborate principle for limit test of Arsenic. Discuss the components of Gutzeit apparatus.
- (b) Write method of preparation and uses of
  - (i) Boric acid
  - (ii) Zinc sulphate
  - (iii) Silver nitrate
- (c) Enlist ideal properties of antacids. Write method of preparation and uses of
  - (i) Calcium carbonate
  - (ii) Magnesium carbonate
- (d) What are hematinic? Write method of preparation and uses of
  - (i) Ferric ammonium citrate
  - (ii) Ferrous gluconate
- (e) Discuss the measurement of radioactivity in detail.

**SECTION C****3. Attempt any one part of the following;****7 x 1 = 7**

- (a) Discuss the sources of impurity with suitable examples.
- (b) Write note on
  - (i) Pharmaceutically acceptable glasses
  - (ii) Inhalants

4. Attempt any **one** part of the following:

7 x 1 = 7

- (a) What for protective agents are used? Discuss it in context with calamine and kaolin.
- (b) Discuss the mechanism of action (any two)
  - (i) Anti-infective agents
  - (ii) Anti-caries agent
  - (iii) Anti-oxidant

5. Attempt any **one** part of the following:

7 x 1 = 7

- (a) Define and classify cathartics. Write method of preparation and uses of
  - (i) Disodium hydrogen phosphate
  - (ii) Magnesium sulfate
- (b) Write note on
  - (i) Expectorant
  - (ii) Respiratory stimulants

6. Attempt any **one** part of the following:

7 x 1 = 7

- (a) List major extracellular and intracellular physiological ions. Discuss the role of sodium ion and chloride ion.
- (b) Write note on
  - (i) Mineral supplements
  - (ii) Replacement therapy

7. Attempt any **one** part of the following:

7 x 1 = 7

- (a) Write in detail clinical applications of radiopharmaceuticals
- (b) Summarize postulates of coordination theory. Enumerate the pharmaceutical importance of chelates.